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Detecting Left Ventricular Systolic Dysfunction in Left Bundle Branch Block Patients Using Electrocardiogram: A Deep Learning Approach with Limited Data

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Abstract

Left ventricular systolic dysfunction (LVSD) is associated with increased mortality and is sometimes reversible when found early. Artificial intelligence (AI)-enabled electrocardiogram (ECG) has emerged as an efficient screening tool for LVSD, but has not been validated in left bundle branch block (LBBB) patients. The clinical significance of developing an AI prediction model for LBBB patients lies in the fact that LBBB can be a cause, consequence, or both of LVSD. This pilot study was designed to develop an AI model for LVSD detection in the LBBB population using a limited dataset. ECG data from 508 patients with sinus rhythm and LBBB were labeled based on an LVSD threshold of 35%. To enhance the performance of a model derived from such a small and skewed dataset, we combined an autoencoder-based anomaly detection model with a convolutional neural network (CNN). We used a lead-wise ensemble technique for the final classification. Experimental results showed an accuracy of 0.81, precision of 0.87, recall of 0.56, and an area under the receiver operating characteristic curve of 0.75 in LVSD prediction among LBBB patients. Despite the limited dataset size, our study findings suggest the potential of deep learning techniques in detecting LVSD in patients with LBBB.

Keywords: left bundle branch block; left ventricular systolic dysfunction; electrocardiogram; autoencoder; convolutional neural network

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1. Introduction

An electrocardiogram (ECG) is a noninvasive test to assess the heart's electrical activity and provides information on cardiac arrhythmia and structural heart disease. A 12-lead ECG records the electrical activity from the body surface in 12 directions on the vertical and horizontal planes. Left bundle branch block (LBBB), which is detected on an ECG when intraventricular conduction via the left bundle is impeded, can be accompanied by left ventricular systolic dysfunction (LVSD) [1,2].

Generally, LVSD is represented by the left ventricular ejection fraction (LVEF), the fraction of blood volume ejected from the left ventricle in systole in relation to end-diastolic volume, which is assessed by echocardiography. Considering the adverse outcomes of LVSD, early recognition of LVSD is important but echocardiography is not readily accessible in real-world clinical settings, especially in asymptomatic individuals. The plasma B-type natriuretic peptide (BNP) level can be used for LVSD detection [3], but it is less useful unless

accompanied by heart failure [4,5]. It is also an invasive test requiring blood sampling, and there is no standardized cut-off value that can be applied across different populations. Compared to the ECG, which is easily repeatable and reproducible, echocardiography and BNP-level measurement require additional resources, leading to increased costs and time requirements. For this reason, research has been conducted to utilize the ECG as an alternative screening tool for LVSD.

Since the 1990s, efforts have been made to utilize ECG for detecting LVSD [6,7]. However, traditional ECG-based diagnostic criteria have limited sensitivity and specificity. With the advent of artificial intelligence (AI) and machine learning techniques, several studies have proposed AI-enabled ECG algorithms for LVSD screening and diagnosis [8–10], demonstrating improved diagnostic performance across diverse populations [11–14]. In addition, research has been conducted in specific clinical subgroups, including intensive care unit patients [15], patients presenting to emergency departments [16], and individuals diagnosed with COVID-19 [17]. Recently, some studies have explored the application of Transformer-based architectures for ECG anomaly detection, showing promising performance in capturing temporal dependencies and handling limited datasets [18,19]. However, most prior AI-ECG models, including these Transformer-based approaches, have not been specifically validated in patients with LBBB.

Despite these advances, most prior studies have focused on general populations and have not been validated specifically in patients with LBBB. The presence of LBBB can alter ECG waveforms significantly, complicating the recognition of LVSD-related features and potentially limiting model performance. Therefore, there remains a critical gap in developing and validating AI models tailored for LVSD prediction in this clinically important subgroup.

To address this gap, we propose a deep learning model designed to predict LVSD specifically in patients with LBBB. Our approach integrates practical strategies, including autoencoder-based pretraining for robust representation learning, a CNN-based classification model for detecting LVSD, and a lead-wise ensemble technique to leverage complementary spatial information from different ECG leads. A pilot experiment was conducted using a dataset from a single tertiary university hospital to evaluate the feasibility and effectiveness of the proposed method in a real-world clinical setting.

2. Materials and Methods

2.1. Data Management

For this study, ECG and LVEF data were provided by a tertiary university hospital. The Institutional Review Board at Samsung Changwon Hospital approved the study protocol (IRB No. 2023-06-005). Data were collected from patients exhibiting sinus rhythm and LBBB on their ECG who had undergone echocardiographic examinations within a 2-month interval from ECG acquisition. LBBB was defined as (1) QRS duration $\geq$ 130 ms; (2) QS or rS in lead V1 and V2; and (3) mid-QRS notching or slurring in 2 or more of leads V1, V2, V5, V6, I, and aVL [20] and confirmed by an electrophysiologist. Each patient's ECG data consists of 10-s 12-lead recordings containing 5000 samples per lead. LVSD was defined as LVEF $\leq$ 35%, while non-LVSD was defined as LVEF $\geq$ 50%. In total, 189 patients were classified as LVSD and 319 as non-LVSD groups. From the entire dataset, 4944 data derived from 412 patients were used for training, while 1152 derived from 96 patients were used as the test set.

2.2. System Framework

Figure 1 shows the framework for building a system to predict LVSD in patients with LBBB. First, the preprocessing steps involve loading ECG data from XML files, applying

a bandpass filter, removing outliers, and scaling the data range processes. Then, only the non-LVSD data from the preprocessed dataset is used to train the anomaly detection autoencoder model. Using the trained autoencoder model, the residual signal between the original data and the reconstructed data are calculated, and these residuals are input into a CNN classification model for training. Finally, the prediction results for the 12-lead data are aggregated using a voting mechanism to derive the final result.

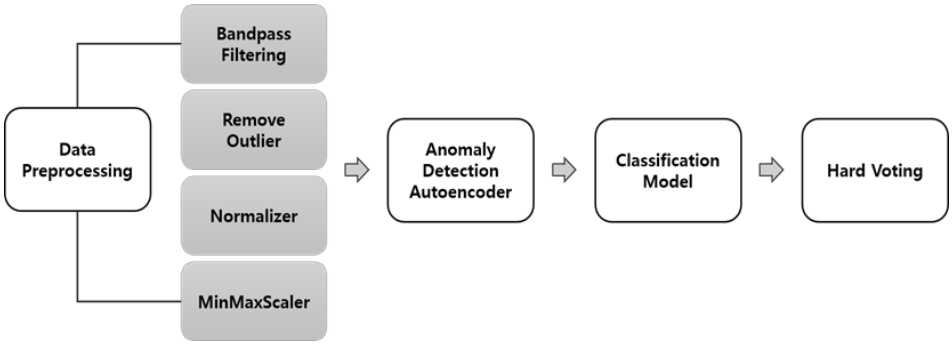


Figure 1. The system framework.

2.3. Preprocessing

2.3.1. Bandpass Filtering

ECG data often contains a significant amount of noise due to various factors such as external interference during measurement, electrode issues, muscle movements, and problems with the measuring equipment. Therefore, noise removal is essential for utilizing ECG data effectively. Bandpass filtering is widely used for noise reduction in ECG datasets, as it is effective at eliminating both high-frequency and low-frequency noise [21,22]. The FFT graph in Figure 2 shows that noise occurs at frequencies above 50 Hz. To retain essential information while removing noise, a bandpass filter allowing only frequencies between 0 and 50Hz to pass was used. Figure 3 shows noise-free ECG and FFT graphs after bandpass filtering, demonstrating that the noise was ideally removed after the bandpass filtering process.

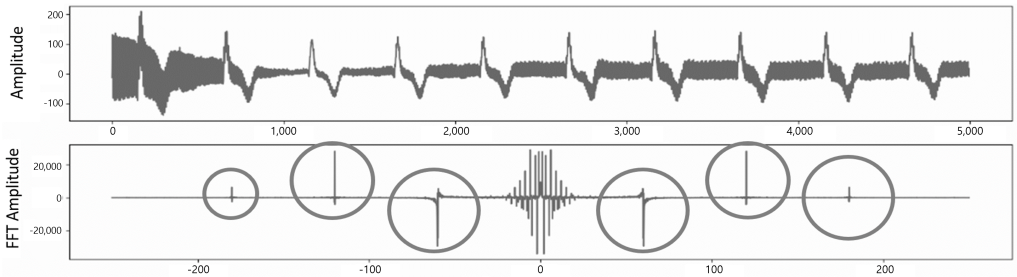


Figure 2. ECG source graph (**top**), and FFT graph (**bottom**).

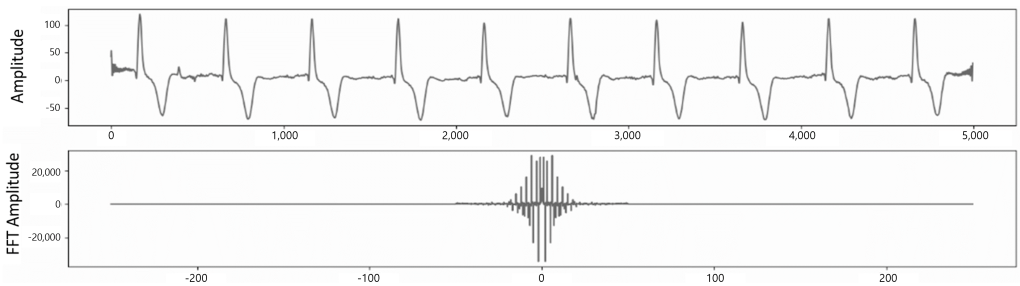


Figure 3. After applying bandpass filtering, ECG graph (**top**) and FFT graph (**bottom**).

2.3.2. Outlier Removal and Feature Normalization

To minimize unnecessary data loss while ensuring the integrity of the dataset, we examined the distribution of each feature. The inspection revealed that nearly all data points fell within the range of −500 to 500, with values beyond this range occurring rarely and likely due to measurement errors or artifacts. Therefore, we applied a consistent threshold by clipping values at −500 and 500 as physiologically plausible limits. Any data point with at least one feature outside this range was considered an outlier and excluded from the training set. As a result, 331 outliers were removed from a total of 4944 samples, yielding 4643 data points for model training.

Following outlier removal, feature scaling was performed using the MinMaxScaler from scikit-learn to rescale all feature values to the range from 0 to 1. In addition, for selected feature sets where vector magnitude uniformity was important, scikit-learn Normalizer (L2 norm) was applied to ensure that the resulting feature vectors have unit length. These preprocessing steps helped stabilize the training process and improve model convergence.

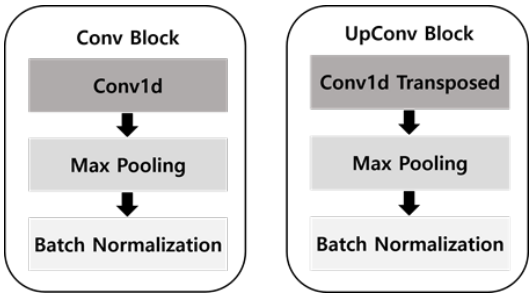
2.4. Autoencoder Anomaly Detection Model

An autoencoder is a self-supervised learning technique with a neural network structure designed to effectively compress data into a lower-dimensional space and reconstruct it with minimized error. It is widely used for various purposes, such as dimensionality reduction, data compression, noise removal, and feature extraction [23]. By leveraging the characteristics of autoencoders, the model can learn only the non-LVSD data patterns and detect anomalies using the reconstruction errors that occur when the trained model reconstructs the input data.

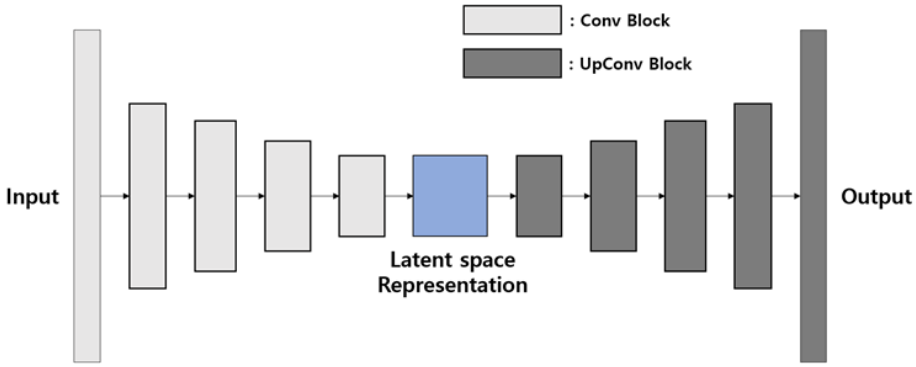
As shown in Figure 4a, the model consists of an encoder and a decoder, each containing four convolutional layers. Figure 4b illustrates the internal structure of the Conv Block and UpConv Block, which are the convolutional layers used in the model.

At the center of the four convolutional layers in both the encoder and decoder lies the latent space. The encoder maps the input data to the latent space, extracting features from the input data and compressing high-dimensional data into low-dimensional data. The latent space refers to the low-dimensional space created after the input data passes through the encoder and is compressed. This space serves as a latent representation that captures the essential features of the input data, which is a core part of the autoencoder. The decoder's role is to reconstruct the input data from the information in the latent space, using the latent representations to reconstruct the input data.

The usefulness of the trained autoencoder model was verified using mean squared error (MSE). This method calculates the reconstruction error between the original signal (input data) and the reconstructed signal (output data). A smaller MSE indicates a smaller difference between the input data and the reconstructed data, implying that the autoencoder more accurately reconstructs the input data. Figure 5 shows a comparison graph of the MSE distribution between non-LVSD and LVSD patients. The difference in the distribution position between the non-LVSD and the LVSD group data shows that the anomaly detection autoencoder was well-trained. This clear separation indicates that the autoencoder successfully learned to reconstruct non-LVSD ECG patterns and produced higher reconstruction errors when processing LVSD data, which differ significantly from non-LVSD training data.



(**a**)



$$\boldsymbol{Residual\ Signal = Input(Original\ Signal) - Output(Restore\ Signal)}$$

(**b**)

Figure 4. The architecture of the Autoencoder Anomaly Detection Model: (**a**) the structure of Conv Block and UpConv Block; (**b**) the structure of the Autoencoder Model.

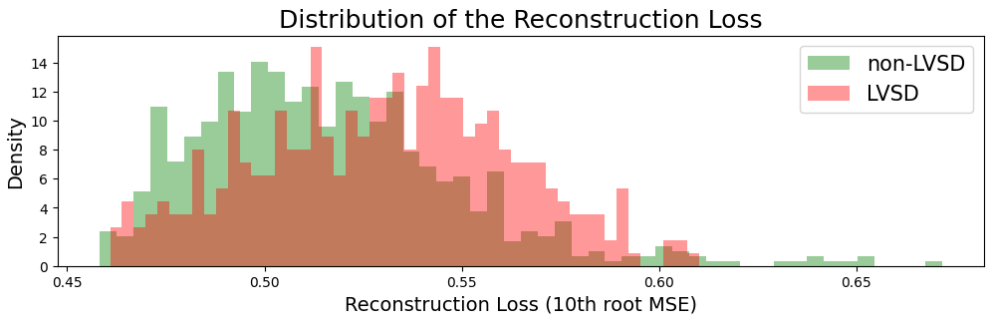


Figure 5. A comparison graph of mean squared error distribution between the original and restored version.

Figure 6 compares the original and autoencoder-reconstructed graphs for a single lead subject. It shows little difference between the original and reconstructed signals in the non-LVSD group, while there is a noticeable difference between the 2 signals in the LVSD group. This indicates that the autoencoder model reconstructs non-LVSD data more accurately than LVSD data, suggesting that the difference in reconstruction errors between LVSD and non-LVSD data can be effectively utilized to identify distinctive features of each signal.

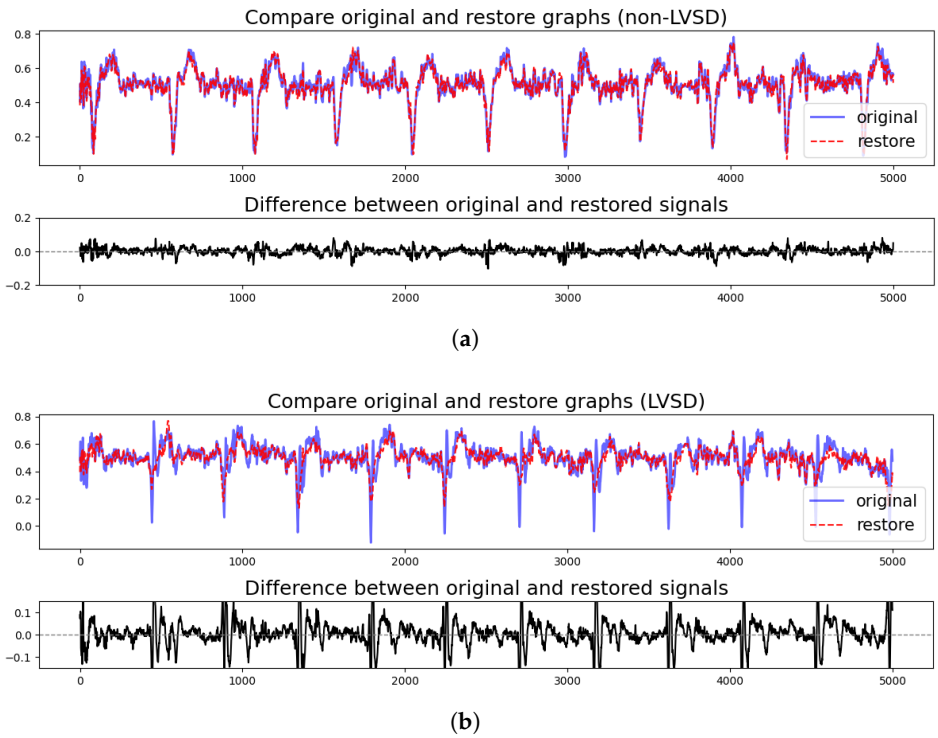


Figure 6. The comparison between the original and restored graphs: (**a**) non-LVSD; (**b**) LVSD.

2.5. Classification Model

Convolutional Neural Network (CNN) is a class of deep learning models particularly effective for analyzing structured data such as images or time-series signals. They automatically extract hierarchical features through layers of convolution and pooling operations, enabling the model to capture both local and global patterns in the data. In recent years, CNN has shown strong performance in biomedical signal analysis, including ECG interpretation. In this study, we used CNN as the classification model to detect LVSD by learning residual ECG patterns characteristic of the non-LVSD group. Specifically, we used the residual signal, obtained by subtracting the reconstructed data from the original data of the pre-trained anomaly detection model, as input for CNN training. Figure 7 shows the CNN classification model structure.

The CNN architecture consisted of three 1D convolutional layers with 8, 16, and 16 filters, respectively, each followed by batch normalization, ReLU activation, and max pooling. The feature maps were then flattened and passed through six fully connected layers with 128, 64, 32, 16, 8, and 2 units, respectively, with dropout layers applied after each dense layer to reduce overfitting.

The model was trained using the Adam optimizer with default parameters, a learning rate of 0.0001, and a batch size of 16. Binary cross-entropy was used as the loss function. To prevent overfitting, early stopping with a patience of 10 epochs based on validation loss was employed. Model performance was monitored using accuracy and loss metrics on the validation dataset.

Given the limited dataset size, this relatively simple architecture yielded better generalization performance than deeper or more complex models.

We conducted Grad-CAM analysis to better understand how the CNN classifier distinguishes LVSD from non-LVSD predictions. Figure 8 presents representative examples of Grad-CAM overlays for one non-LVSD and one LVSD case. In the non-LVSD example, attention was sparse and localized, while in the LVSD example, attention was broadly distributed across the entire signal.

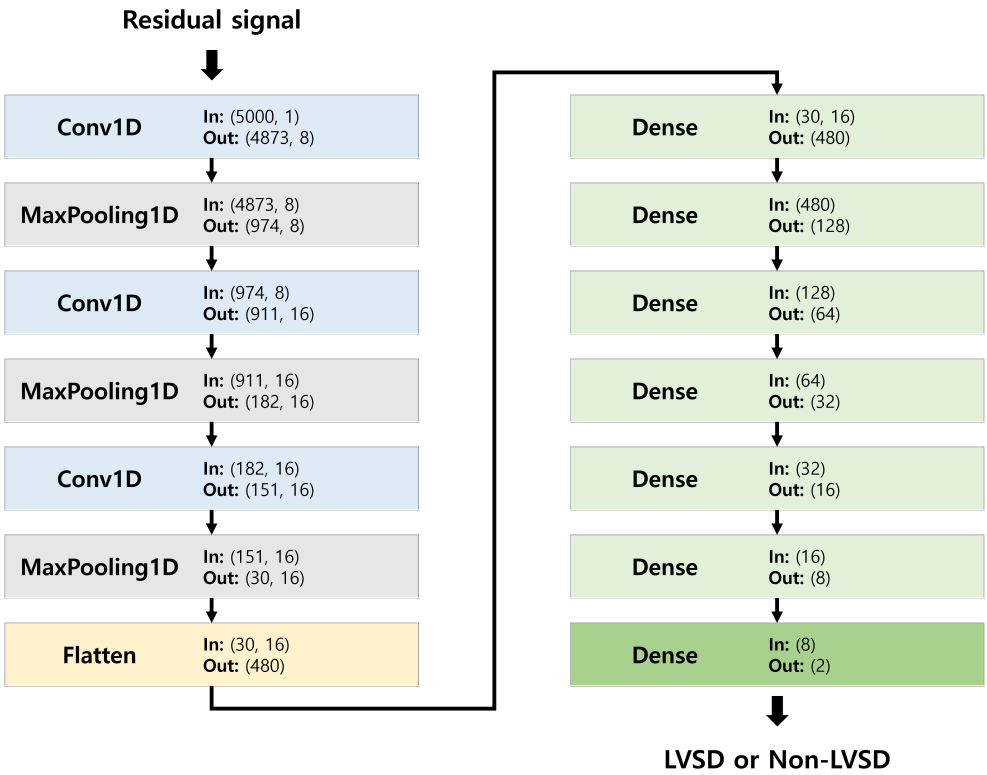


Figure 7. Proposed 1D CNN classification model architecture.

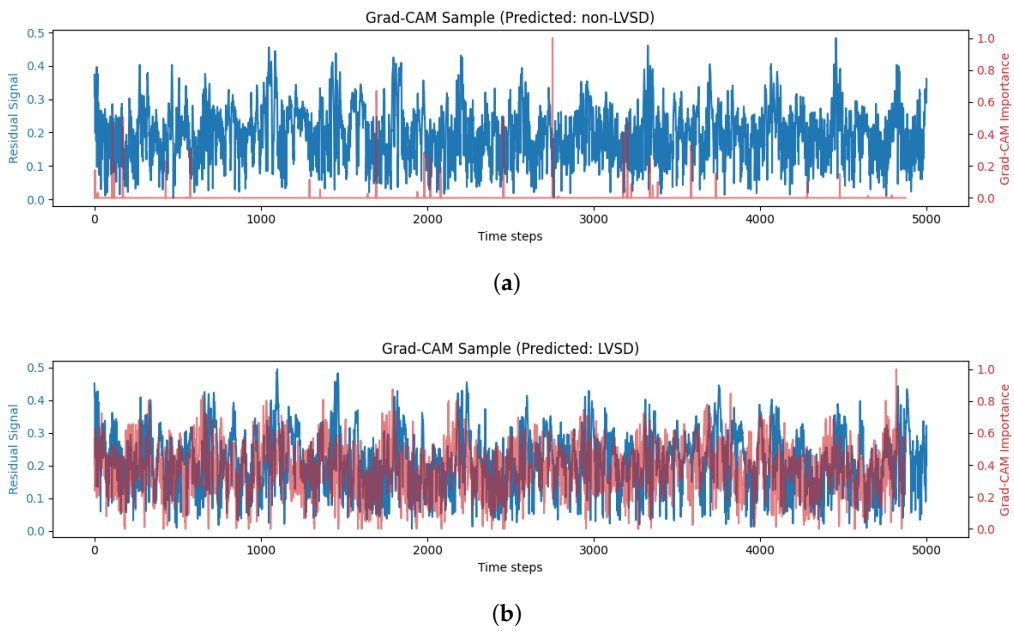


Figure 8. Representative Grad-CAM overlays for individual test cases: (**a**) non-LVSD prediction; (**b**) LVSD prediction.

To confirm this pattern at the dataset level, we computed the mean and standard deviation Grad-CAM profiles for all test predictions (Figure 9). The non-LVSD group showed consistently low attention with minimal variability, whereas the LVSD group exhibited elevated global attention with greater variability, suggesting that the model generally relies on global residual abnormalities when predicting LVSD.

These results indicate that the model adopts distinct attention strategies: minimal, localized attention for non-LVSD predictions, and diffuse, global attention for LVSD predictions, providing explainable insight into the model's decision process.

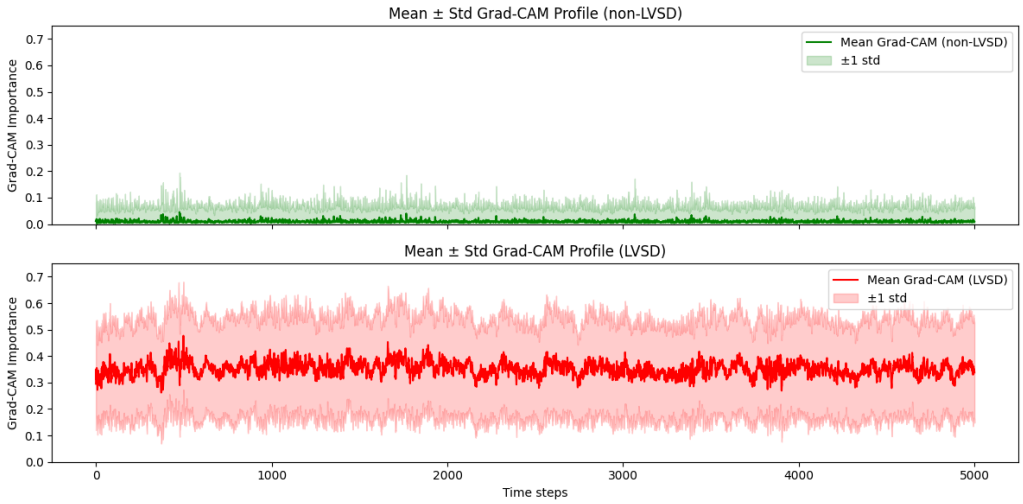


Figure 9. Mean ± 1 std Grad-CAM profiles for test predictions: (**top**) non-LVSD, (**bottom**) LVSD.

2.6. Lead-Wise Ensemble

Ensemble learning is a powerful machine learning strategy that combines the outputs of multiple individual models to produce more accurate and reliable predictions than any single model alone. By aggregating diverse perspectives or decision boundaries, ensemble methods help reduce variance, mitigate overfitting, and improve generalization performance-particularly in settings with noisy or heterogeneous data.

In this study, we applied an ensemble approach tailored to multi-lead ECG data, where each lead provides complementary diagnostic information for detecting LVSD in patients with LBBB. To enhance recognition performance, the final classification results were derived using a lead-wise ensemble technique. Each ECG lead captures cardiac activity from a different spatial perspective, and certain LVSD-related features may be more pronounced in specific leads. By aggregating predictions across multiple lead-specific classifiers, the ensemble approach leverages complementary information and mitigates the risk of overfitting to any single lead.

As illustrated in Figure 10, the lead-wise ensemble operates by first obtaining predictions from individual CNN classifiers, each trained on a specific ECG lead. These lead-specific prediction results are then input into a final voting-based classifier, which determines the overall classification outcome. This structure allows the model to integrate diverse diagnostic signals across leads in a systematic and robust manner.

In this study, we adopted hard voting, where each classifier casts a discrete vote for the predicted class, and the final prediction is determined by majority rule. We chose hard voting over soft voting to increase robustness against potential noise or abnormal signals in specific leads. In soft voting, classifiers output probability scores, which are then averaged to determine the final class. However, if one or more leads produce abnormally high or low probabilities due to artifacts or outliers, the final decision may be disproportionately influenced. Hard voting avoids this issue by treating each classifier's decision equally, thus enhancing the stability of the ensemble.

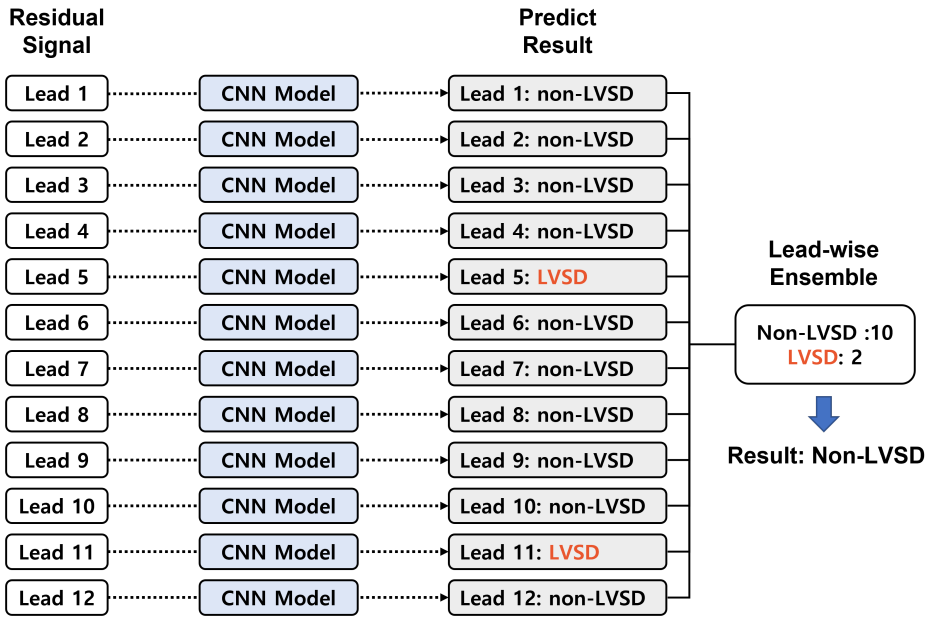


Figure 10. A lead-wise ensemble example.

3. Results and Discussion

3.1. Validation of an Existing Model

In 2022, a study trained a 2D CNN model to detect patients with an LVEF below 40% and provided a demonstration site to showcase the model [11]. In 2024, another study validated this pre-trained model using an external dataset [12]. We also validated the demo model using our dataset. The model's output is the probability of LVSD, expressed as a continuous variable between 0 and 1. Therefore, we used Youden's J index to find the optimal cut-off. The cut-off value where Youden's J is maximized represents the balance between sensitivity and specificity and is considered the optimal threshold for test performance [24]. The test results shown in Figure 11 indicate that the optimal cutoff value for the dataset used in this study is 0.858, as represented by the red line.

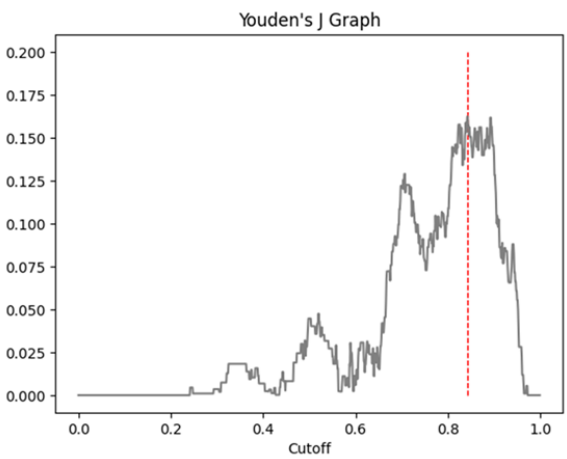


Figure 11. Youden's J cut-off graph.

The prediction results using the pre-trained model for the 100 subjects in the overall lead test were as follows: an accuracy of 0.6, a precision of 0.62, a recall of 0.2, an F1-score of 0.28, and an AUC of 0.56. These results suggest that the pre-trained model does not perform well on datasets with LBBB.

In addition to validating the pre-trained AI model, we conducted a reference analysis using a simple rule-based criterion based on QRS duration. Specifically, we compared prediction performance at various QRS duration thresholds to examine its utility as a basic clinical benchmark for LVSD prediction in patients with LBBB. To determine the optimal threshold, we also plotted Youden's J index across different cut-off values, as shown in Supplementary Figure S1. Detailed accuracy by threshold is summarized in Table S1.

3.2. CNN Model Training: Single-Lead Results

The CNN model was trained using the residuals between the reconstructed signals obtained from the autoencoder model and the original signal values. The training results showed an accuracy of 0.74 for the 12 single-lead test datasets from 100 subjects. The test results showed an accuracy of 0.74, a precision of 0.7, a recall of 0.48, an F1-score of 0.58, and an AUC of 0.69.

Table 1 presents the detailed performance metrics for each single-lead prediction model. The accuracy, AUC, F1-score, precision, and recall were evaluated for each of the 12 single leads. The results indicate that while there are differences in performance metrics among the leads, the variations are minimal, and thus, no significant differences were observed.

Table 1. Single lead performance metrics.

Lead	Accuracy	Precision	Recall	F1-Score	AUC
I	0.75	0.8	0.43	0.56	0.68
II	0.76	0.69	0.65	0.67	0.74
III	0.77	0.75	0.57	0.65	0.73
aVR	0.74	0.7	0.51	0.59	0.69
aVL	0.76	0.78	0.49	0.6	0.7
aVF	0.72	0.66	0.51	0.58	0.68
V1	0.74	0.92	0.34	0.5	0.66
V2	0.79	0.92	0.44	0.59	0.71
V3	0.76	0.85	0.41	0.55	0.68
V4	0.73	0.68	0.43	0.53	0.66
V5	0.72	0.62	0.5	0.55	0.67
V6	0.72	0.65	0.49	0.55	0.67

3.3. Lead-Wise Ensemble Prediction: Multi-Lead Results

The overall lead test results for the 100 subjects, using the lead-wise ensemble method, showed an accuracy of 0.81, a precision of 0.87, a recall of 0.56, an F1-score of 0.68, and an AUC of 0.75. The lead-wise ensemble prediction results demonstrated an overall improvement compared to the single-lead CNN model prediction results. Table 2 shows the results of summarizing the experimental results.

Table 2. Comparison of experimental results.

Experimental Cases	Accuracy	Precision	Recall	F1-Score	AUC
Existing Model	0.6	0.62	0.2	0.28	0.56
Single Lead (CNN Model)	0.74	0.7	0.48	0.58	0.69
Multi Lead (Lead-Wise Ensemble)	0.81	0.87	0.56	0.68	0.75

3.4. Discussion

Identification of LVSD has clinical significance because, even if asymptomatic, it is associated with increased mortality and morbidities. On the other hand, there is an opportunity to prevent the progression of LVSD with timely medical and/or device therapy [25,26].

Conventional tools for assessing LVEF, including echocardiography, cardiac magnetic resonance imaging, and cardiac computed tomographic angiography, are resource-intensive and time-consuming when used for screening purposes in real-world clinical settings. In this context, recent AI models using the ECG, an inexpensive and readily available tool even in resource-limited environments, have attracted great attention, showing high performance in LVSD detection.

LBBB is well known to be associated with LVSD, whether it is the cause or consequence of early cardiac structural changes. However, pre-existing models have not been validated for patients with LBBB. As shown in the Section 3.1, an existing open-source algorithm [12] did not perform well with our LBBB data. Thus, the development of advanced AI models is necessary to detect LVSD in the LBBB population.

This pilot study focused on exploring the feasibility of an AI-enabled ECG approach rather than developing a high-performance model, considering the small dataset obtained from a single institution. Previous studies on ECG-based heart disease prediction primarily utilized a single CNN model or methods that combined CNN with other AI techniques such as U-Net, RNN, and LSTM models, demonstrating relatively high performance. However, considering the limitations of the dataset size in this study, the proposed approach that combines a CNN model with an autoencoder-based anomaly detection model outperformed these conventional methods. Although there are limitations in model performance and dataset size, the findings of this study suggest the potential applicability of deep learning techniques in detecting LVSD among patients with LBBB. There would be a merit to apply an AI algorithm for LBBB detection to enhance the efficiency of LVSD prediction model even in non-LBBB subjects. Even though we used an LVEF $\leq$ 35%, a fine-tuned AI model for detecting structural and/or functional abnormality in the early phase, including mild reduction in LVEF, will carry more clinical relevance as it can allow timely intervention.

However, certain limitations must be acknowledged. The small dataset size may have impacted the model's generalizability, and further validation with larger, more diverse datasets is necessary. Additionally, due to the small sample size, we used a single train-test split to ensure strict separation between training and validation sets; however, we acknowledge that k-fold cross-validation or bootstrapping could provide more stable and generalizable estimates, and we plan to adopt such methodologies in future studies. Furthermore, this study did not test generalization to out-of-distribution (OOD) data, i.e., data from populations or acquisition settings differing from the training set. Given the importance of OOD robustness for clinical deployment, future work will incorporate methodologies for OOD evaluation and enhancement as discussed in recent literature [27,28].

While a CNN-based architecture was chosen for its effectiveness on small datasets and structured time-series data such as ECG, Transformer architectures have demonstrated excellent performance in many domains but typically require larger datasets for training and generalization [29]. We did not include a comparative analysis with Transformer models due to dataset size limitations, but future studies will explore and benchmark these architectures when sufficient data become available. Additionally, while deep learning models can detect patterns in ECG signals, the interpretability of these models remains a challenge.

To improve interpretability, we performed Grad-CAM analysis and observed distinct attention patterns for LVSD and non-LVSD predictions, offering insight into the model's decision process. Clinicians require transparent and explainable AI-based diagnostic tools to ensure trust and seamless integration into clinical practice. Addressing these issues will be critical for future advancements in AI-driven cardiovascular assessments. To improve the model's robustness and clinical applicability, we are currently in the process of acquiring multi-center data.

4. Conclusions

This study developed and validated a tailored deep learning model that predicts LVSD from 12-lead electrocardiograms in patients with LBBB. As summarized in Table 2, the proposed lead-wise ensemble CNN achieved an AUC of 0.75 and an F1-score of 0.68, demonstrating improved performance compared to the existing publicly available LVSD detection model, which was not specifically designed for patients with LBBB.

These results confirm that LVSD can be detected with clinically useful accuracy even when only limited ECG data are available, addressing a key gap in previous research that rarely focused on the LBBB population. By integrating explainable saliency mapping, the model also offers insight into lead-specific ECG patterns, thereby improving clinical interpretability. Collectively, our findings highlight the potential of AI-driven diagnostic systems to enable earlier detection, more confident clinical decision-making, and ultimately better outcomes for patients with suspected LVSD who present with LBBB.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/app15158384/s1, Figure S1: Determining the optimal QRS duration cut-off for LVSD prediction using Youden's index; Table S1: Summary of LVSD prediction performance at various QRS duration cut-off values.

Author Contributions: Conceptualization, C.K., H.B.G. and J.S.; methodology, C.K.; software, C.K.; validation, C.K., H.B.G. and J.S.; formal analysis, C.K.; investigation, C.K.; resources, H.B.G. and J.S.; data curation, C.K. and H.B.G.; writing—original draft preparation, C.K. and H.B.G.; writing—review and editing, H.B.G. and J.S.; visualization, C.K.; supervision, H.B.G. and J.S.; project administration, H.B.G.; and funding acquisition, J.S. All authors have read and agreed to the published version of the manuscript.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
BNP	B-type natriuretic peptide
CNN	Convolutional neural network
ECG	Electrocardiogram
LBBB	Left bundle branch block
LVSD	Left ventricular systolic dysfunction
MSE	Mean squared error

References

1. Lau, E.W.; Bonnemeier, H.; Baldauf, B. Left bundle branch block—Innocent bystander, silent menace, or both. *Heart Rhythm* **2024**, *22*, e229–e236. [CrossRef]
2. Tan, N.Y.; Witt, C.M.; Oh, J.K.; Cha, Y.M. Left bundle branch block: Current and future perspectives. *Circ. Arrhythm. Electrophysiol.* **2020**, *13*, e008239. [CrossRef]

3. Costello-Boerrigter, L.C.; Boerrigter, G.; Redfield, M.M.; Rodeheffer, R.J.; Urban, L.H.; Mahoney, D.W.; Jacobsen, S.J.; Heublein, D.M.; Burnett, J.C. Amino-terminal pro-B-type natriuretic peptide and B-type natriuretic peptide in the general community: Determinants and detection of left ventricular dysfunction. *J. Am. Coll. Cardiol.* **2006**, *47*, 345–353. [CrossRef]
4. Cho, J.; Lee, B.; Kwon, J.M.; Lee, Y.; Park, H.; Oh, B.H.; Jeon, K.H.; Park, J.; Kim, K.H. Artificial intelligence algorithm for screening heart failure with reduced ejection fraction using electrocardiography. *ASAIO J.* **2021**, *67*, 314–321. [CrossRef] [PubMed]
5. Attia, Z.I.; Kapa, S.; Lopez-Jimenez, F.; McKie, P.M.; Ladewig, D.J.; Satam, G.; Pellikka, P.A.; Enriquez-Sarano, M.; Noseworthy, P.A.; Munger, T.M.; et al. Screening for cardiac contractile dysfunction using an artificial intelligence–enabled electrocardiogram. *Nat. Med.* **2019**, *25*, 70–74. [CrossRef] [PubMed]
6. McDonagh, T.; Robb, S.; Murdoch, D.; Morton, J.; Ford, I.; Morrison, C.; Tunstall-Pedoe, H.; McMurray, J.; Dargie, H. Biochemical detection of left-ventricular systolic dysfunction. *Lancet* **1998**, *351*, 9–13. [CrossRef]
7. Davie, A.; Francis, C.; Love, M.; Caruana, L.; Starkey, I.; Shaw, T.; Sutherland, G.; McMurray, J. Value of the electrocardiogram in identifying heart failure due to left ventricular systolic dysfunction. *BMJ* **1996**, *312*, 222. [CrossRef]
8. Choi, J.; Lee, S.; Chang, M.; Lee, Y.; Oh, G.C.; Lee, H.Y. Deep learning of ECG waveforms for diagnosis of heart failure with a reduced left ventricular ejection fraction. *Sci. Rep.* **2022**, *12*, 14235. [CrossRef]
9. Potter, E.L.; Rodrigues, C.H.; Ascher, D.B.; Abhayaratna, W.P.; Sengupta, P.P.; Marwick, T.H. Machine learning of ECG waveforms to improve selection for testing for asymptomatic left ventricular dysfunction. *Cardiovasc. Imaging* **2021**, *14*, 1904–1915. [CrossRef]
10. Yao, X.; Rushlow, D.R.; Inselman, J.W.; McCoy, R.G.; Thacher, T.D.; Behnken, E.M.; Bernard, M.E.; Rosas, S.L.; Akfaly, A.; Misra, A.; et al. Artificial intelligence–enabled electrocardiograms for identification of patients with low ejection fraction: A pragmatic, randomized clinical trial. *Nat. Med.* **2021**, *27*, 815–819. [CrossRef] [PubMed]
11. Yagi, R.; Goto, S.; Katsumata, Y.; MacRae, C.A.; Deo, R.C. Importance of external validation and subgroup analysis of artificial intelligence in the detection of low ejection fraction from electrocardiograms. *Eur. Heart J.-Digit. Health* **2022**, *3*, 654–657. [CrossRef] [PubMed]
12. König, S.; Hohenstein, S.; Nitsche, A.; Pellissier, V.; Leiner, J.; Stellmacher, L.; Hindricks, G.; Bollmann, A. Artificial intelligence-based identification of left ventricular systolic dysfunction from 12-lead electrocardiograms: External validation and advanced application of an existing model. *Eur. Heart J.-Digit. Health* **2024**, *5*, 144–151. [CrossRef]
13. Kashou, A.H.; Medina-Inojosa, J.R.; Noseworthy, P.A.; Rodeheffer, R.J.; Lopez-Jimenez, F.; Attia, I.Z.; Kapa, S.; Scott, C.G.; Lee, A.T.; Friedman, P.A.; et al. Artificial intelligence–augmented electrocardiogram detection of left ventricular systolic dysfunction in the general population. *Mayo Clin. Proc.* **2021**, *96*, 2576–2586. [CrossRef]
14. Mondo, C.; Attia, Z.; Benavente, E.; Friedman, P.; Noseworthy, P.; Kapa, P.; Ingabire, P.; Semanda, S.; Perel, P.; Lopez-Jimenez, F. External validation of an electrocardiography artificial intelligence-generated algorithm to detect left ventricular systolic function in a general cardiac clinic in Uganda. *Eur. Heart J.* **2020**, *41*, ehaa946.1013. [CrossRef]
15. Jentzer, J.C.; Kashou, A.H.; Attia, Z.I.; Lopez-Jimenez, F.; Kapa, S.; Friedman, P.A.; Noseworthy, P.A. Left ventricular systolic dysfunction identification using artificial intelligence-augmented electrocardiogram in cardiac intensive care unit patients. *Int. J. Cardiol.* **2021**, *326*, 114–123. [CrossRef]
16. Adedinsewo, D.; Carter, R.E.; Attia, Z.; Johnson, P.; Kashou, A.H.; Dugan, J.L.; Albus, M.; Sheele, J.M.; Bellolio, F.; Friedman, P.A.; et al. Artificial intelligence-enabled ECG algorithm to identify patients with left ventricular systolic dysfunction presenting to the emergency department with dyspnea. *Circ. Arrhythm. Electrophysiol.* **2020**, *13*, e008437. [CrossRef]
17. Attia, Z.I.; Kapa, S.; Noseworthy, P.A.; Lopez-Jimenez, F.; Friedman, P.A. Artificial intelligence ECG to detect left ventricular dysfunction in COVID-19: A case series. *Mayo Clin. Proc.* **2020**, *95*, 2464–2466. [CrossRef]
18. Zhou, Y.; Yang, Y.; Gan, J.; Li, X.; Yuan, J.; Zhao, W. Multi-scale Masked Autoencoder for Electrocardiogram Anomaly Detection. *arXiv* **2025**, arXiv:2502.05494.
19. Alghieth, M. DeepECG-Net: A hybrid transformer-based deep learning model for real-time ECG anomaly detection. *Sci. Rep.* **2025**, *15*, 20714. [CrossRef] [PubMed]
20. Strauss, D.G.; Selvester, R.H.; Wagner, G.S. Defining left bundle branch block in the era of cardiac resynchronization therapy. *Am. J. Cardiol.* **2011**, *107*, 927–934. [CrossRef] [PubMed]
21. Xiao, Q.; Lee, K.; Mokhtar, S.A.; Ismail, I.; Pauzi, A.L.b.M.; Zhang, Q.; Lim, P.Y. Deep Learning-Based ECG Arrhythmia Classification: A Systematic Review. *Appl. Sci.* **2023**, *13*, 4964. [CrossRef]
22. Narotamo, H.; Dias, M.; Santos, R.; Carreiro, A.V.; Gamboa, H.; Silveira, M. Deep learning for ECG classification: A comparative study of 1D and 2D representations and multimodal fusion approaches. *Biomed. Signal Process. Control* **2024**, *93*, 106141. [CrossRef]
23. Kingma, D.P.; Welling, M. Auto-encoding variational bayes. *arXiv* **2013**, arXiv:1312.6114.
24. Thiele, C.; Hirschfeld, G. cutpointr: Improved estimation and validation of optimal cutpoints in R. *J. Stat. Softw.* **2021**, *98*, 1–27. [CrossRef]
25. Rickard, J.; Michtalik, H.; Sharma, R.; Berger, Z.; Iyoha, E.; Green, A.R.; Haq, N.; Robinson, K.A. Predictors of response to cardiac resynchronization therapy: A systematic review. *Int. J. Cardiol.* **2016**, *225*, 345–352. [CrossRef] [PubMed]

26. McMurray, J.J.; Packer, M.; Desai, A.S.; Gong, J.; Lefkowitz, M.P.; Rizkala, A.R.; Rouleau, J.L.; Shi, V.C.; Solomon, S.D.; Swedberg, K.; et al. Angiotensin–neprilysin inhibition versus enalapril in heart failure. *N. Engl. J. Med.* **2014**, *371*, 993–1004. [CrossRef] [PubMed]
27. Ye, N.; Li, K.; Bai, H.; Yu, R.; Hong, L.; Zhou, F.; Li, Z.; Zhu, J. Ood-bench: Quantifying and understanding two dimensions of out-of-distribution generalization. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, New Orleans, LA, USA, 18–24 June 2022; pp. 7947–7958.
28. Ye, N.; Zeng, Z.; Zhou, J.; Zhu, L.; Duan, Y.; Wu, Y.; Wu, J.; Zeng, H.; Gu, Q.; Wang, X.; et al. Ood-control: Generalizing control in unseen environments. *IEEE Trans. Pattern Anal. Mach. Intell.* **2024**, *46*, 7421–7433. [CrossRef]
29. Zhu, H.; Chen, B.; Yang, C. Understanding why vit trains badly on small datasets: An intuitive perspective. *arXiv* **2023**, arXiv:2302.03751. [CrossRef]

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